

Kiarash Rezaei

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Research Interests

Statistical Foundations of Trustworthy Machine Learning; Uncertainty Quantification and Reliability Guarantees; Generalization under Distribution Shift and Dependence; Robust and Continual Learning in Autonomous Networked Systems.

Education

- 2025 – Present **Chalmers University of Technology, Department of Electrical Engineering** – Gothenburg, Sweden
PhD in Electrical Engineering
Supervisor: Prof. Paolo Monti, Co-supervisor: Dr. Carlos Natalino
- 2021 – 2024 **Polytechnic University of Milan (PoliMi), Department of Electronics, Information and Bioengineering** – Milan, Italy
MSc in Telecommunication Engineering (Track: Signal and Data Analysis)
Thesis: *Continuous IMU-MEMS Self-Calibration Process by Means of Tiny Neural Networks*
Supervisor: Prof. Luca Barletta
GPA: 106/110
- 2014 – 2020 **Kharazmi University of Tehran (KhU), Department of Mathematics and Computer Science** – Tehran, Iran
BSc in Computer Science
Thesis: *Automatic Architecture Design of CNNs using Genetic Algorithm and Reinforcement Learning (MetaQNN)*
Supervisor: Prof. Keivan Borna

Research experience

- Feb 2025 – Present **Chalmers University of Technology, Communication, Antennas and Optical Networks (CAOS) Division** – Gothenburg, Sweden
Research Assistant
- Working on Trustworthy AI algorithms for Autonomous Networks.

- Nov 2023 – Jul 2024 **STMicroelectronics** – Milan, Italy
AI Researcher
- Designed and developed a scalable, end-to-end pipeline for continuous self-calibration of next-generation MEMS inertial measurement sensors, integrating intelligent sensor processing units to enhance performance and reliability.
 - Implemented a deep learning-based module utilizing Edge AI and Quantization-Aware Training (QAT), achieving a significant 70% reduction in calibration loss, enabling efficient sensor operation in resource-constrained environments.
 - Published 3 scientific papers detailing the innovative methods and results derived from this work.
- Oct 2023 – Nov 2023 **Envision** – Hague, Netherlands
Computer Vision Intern
- Conducted evaluation of multi-modal models trained on ego-centric datasets, benchmarking performance across diverse metrics.
- Apr 2022 – Jul 2023 **PoliMi Data Science Association (PMDS)** – Milan, Italy
Machine Learning Researcher
- Collaborated with Sares Miramondi Co. on a project of Anomaly Detection for production lines.
 - Analyzed the results of the sentiment analysis to identify trends and insights related to public sentiment toward COVID-19.

Publications

- 2026 **Policy-driven Conformal Prediction for Trustworthy QoT Estimation**
Rezaei, K., Ayoub, O., Monti, P., Natalino, C.
- Presented at *OFC 2026*.
- 2026 **Human-Grounded Evaluation of Large Language Models for Optical Network Automation**
Rezaei, K., Ayoub, O., Monti, P., Natalino, C.
- Invited paper at *ICTON 2026*.
- 2026 **Bridging the Trust Gap in AI-Driven Optical Networks with Structured Explainability**
Rezaei, K., Ayoub, O., Natalino, C., Monti, P.
- Invited paper at *OECC 2026*.
- 2025 **Generative Explainability for Next-Generation Networks: LLM-Augmented XAI with Mutual Feature Interactions**
Rezaei, K., Ayoub, O., Troia, S., Lelli, F., Monti, P., Natalino, C.
- Presented at *IEEE WiMob 2025*.

- 2024 **IMU Self-Calibration by Means of Quantization-Aware and Memory-Parsimonious Neural-Networks**
 Cardoni, M., Pau, D. P., Rezaei, K., & Mura, C.
 - Published at *Journal of Electronics*.
- 2024 **IMU User Transparent Tiny Neural Self-Calibration**
 Cardoni, M., Pau, D. P., Rezaei, K.
 - Presented at *IEEE RTSI 2024*.
- 2024 **Continuous MEMS Self-Calibration Process by Means of Tiny Neural Networks**
 Cardoni, M., Pau, D. P., Rezaei, K.
 - Presented at *STMicroelectronics TechWeek 2024*.
 - Submitted as an innovation proposal.
- 2019 **An Introduction to Convolutional Neural Networks & Applications**
 Rezaei, K., Zamani, S.
 Presented at *CICIS 2019*.

Industry Experience

- Oct 2024 – Jan 2025 **AGap2 - Rina** – Milan, Italy
 Generative AI Engineer
 - Collaborated with Rina to develop LLM-based systems for automated assessment of inspection report completeness and coherence.
 - Integrated models into the digital reporting platform, accelerating deployment and delivering a functional MVP within three months.
- Feb 2023 – Feb 2024 **DataLobster** – Paris, France
 Data Scientist
 - Optimized signal processing and ML algorithms for streaming data, boosting performance by 20% and lowering latency.
 - Developed anomaly detection models with explainable AI, enabling insightful root-cause analysis.

Teaching and Supervision Experience

- Aug 2025 – Present **Teaching Assistant, EEN060/EEN065: Applied Object-Oriented Programming, Chalmers University of Technology**
 - Designing examinations and leading exercise and laboratory sessions in Python, guiding students in mastering object-oriented programming concepts and practical implementation.

Jan 2026 – **Supervisor, EENX16, Chalmers University of Technology**
Present - Supervising a bachelor's thesis on autonomous AI systems: "Autonomous Knowledge Agent: From Query to Report without Human Guidance" (EENX16).

Professional memberships

2025 – Present IEEE Student Member